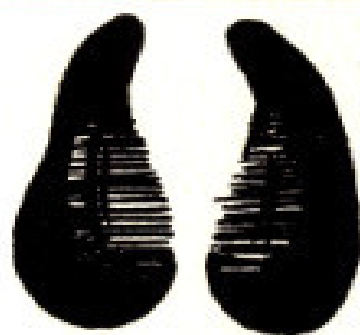




Wild animals *of the Rockies*



MT. SHEEP



COUGAR



BLACK BEAR

by James Gilligan



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MT. SHEEP



COUGAR



BLACK BEAR

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Title: Wild Animals of the Rockies

Author: James P. Gilligan

Release date: July 10, 2019 [eBook #59885]

Language: English

Other information and formats: www.gutenberg.org/ebooks/59885

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*** START OF THE PROJECT GUTENBERG EBOOK WILD ANIMALS
OF THE ROCKIES ***

wild animals of the ROCKIES

**with a list of mammals found
in Rocky Mountain National Park**

By James Gilligan

Additional copies of this booklet may be obtained by writing to the author at Boise Junior College, Boise, Idaho

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PREFACE

Among the frequent questions by visitors to the Rocky Mountain region are those pertaining to the wild animals. What animals are found here? How can they be identified? What are the wildlife problems of the high country? These are common queries which this booklet attempts to answer.

The author, a former ranger-naturalist in Rocky Mountain National Park, discovered through a visitor study in 1948 that a predominating interest of vacationers was in the wildlife of the area. Therefore, the writing has been limited to brief descriptions of the four-footed animals as they are seen in nature, with some explanation of their habits and habitat so they may be more readily located. The cover "tracks" and identification plates further this intent.

Many find it difficult to understand why they cannot see "more" animals in this rugged country. These animals are wild in the strictest sense. Many are nocturnal in habit, hiding during daylight, and others must be approached very cautiously. One satisfactory method of observing wildlife is to select a "spot" off the beaten trails and sit quietly for several hours, allowing animal life to move about in a normal manner.

A secondary purpose of this booklet is to provide a check list of all mammals known to use the National Park. Not all species listed have been collected in the area. It is hoped this will be a start toward providing an accurate, more substantial, and growing list of mammals for the Park. The scientific names of the 50 species given conform to all revisions to date. Those interested in a comprehensive discussion of individual mammals should consult such publications as Warren's **Mammals of Colorado**, or Cahalane's **Mammals of North America**.

An effort has been made to reduce and simplify the many common names attached to certain mammals. The most representative, and yet accurate name, has been selected for each animal for use throughout its entire area of distribution. For example, there is a large group of ground squirrels

(Callospermophilus) inhabiting most western states which closely resemble one another in external features. The variety of common names given these squirrels (due to differences in locality or in minute external characters) is highly perplexing to the average person. Therefore the name golden-mantled ground squirrel, by which most of this group is known in far western states, is given for the group representative in north central Colorado, formerly 6 known as the Say's ground squirrel. Similar methods have been followed in limiting the names of other mammals. It is suggested that those interested in wildlife adopt one common name for each similar group of animals in an effort to standardize terminology for the multitude.

Dr. William H. Burt, Curator of Mammals, University of Michigan Museum of Zoology, has reviewed the manuscript and made many helpful suggestions. Miss Diana Wiltse, of the Ann Arbor Press, designed the cover and identification pages. I am also grateful for files of information and many photographic cuts furnished by the National Park Service. Unless otherwise indicated, photos were provided through the courtesy of Nature Magazine.

WILDLIFE OF THE LAST HUNDRED YEARS

Before the arrival of settlers in the mountain valleys of north-central Colorado in 1860, only the Ute and Arapaho Indians of the region and a few adventuresome white trappers knew the wildlife then so abundant throughout the area. They alone had observed the thousands of elk and small groups of mountain bison (buffalo) grazing the alpine meadows in the summer. Plentiful herds of mule deer roamed the valleys and forests and hundreds of flocks of wild mountain sheep fed on tufts of grass and flowers on rugged mountain slopes above timberline. Even an infrequent moose wandering down from his more northerly habitat could be seen in the lower wet meadows.

Along with these wonderful groups of hoofed animals lived the predatory, or carnivorous, animals. The powerful and vicious wolverines were common in the heavy forests of the high mountains, as were huge grizzly bears, occasional timber wolves, and cougars. The Canada lynx and mountain bobcat preyed on animals their size and smaller in the pine and spruce timber, while the red fox and rare gray fox were effective squirrel, mouse, and rabbit catchers. Pine martens chased small rodents through deeper forested regions, and they in turn were preyed on the larger carnivores. Numerous otter, mink, and weasel played in and along rushing mountain streams and clear lakes, feeding on a great variety of aquatic animals, fish, and small meadow rodents.

This scene might have remained relatively undisturbed had not white man arrived in the region. His rapid settlement and use of the area after 1870 had a startling and widespread effect on the wildlife populations in the next fifty years. Had he been satisfied to develop a small portion of the mountain country and take only sufficient food and clothing materials as they were needed from the wild species, the story might have been different. Instead, the unprecedented mountain scenery, climate, and animals attracted scores of vacationists, sport hunters, trappers, and market hunters, all anxious to profit from the newly-discovered virgin territory.

The great fur demand in St. Louis and Denver at this time attracted scores of trappers; they diligently pursued the valuable fur-bearers until, in 1915, the otters and wolverines were practically extinct in the Rocky Mountain regions of Colorado, and remain in that condition today. A few have been seen recently in the state, but they are exceedingly rare. Beaver were heavily trapped even before the settlers arrived. Mink, marten, and red fox also 8 reached a very low ebb in population in the early 1900's.

The "sportsmen" and market hunters were taking an even greater toll with their systematic slaughter of big game animals. The small herds of mountain bison, as well as any sign of transient moose, had disappeared by 1865. Deer and elk were so plentiful and easy to kill that wagon loads of their meat were hauled from the mountains to Denver markets where they were sold for as little as four cents a pound. Probably the easiest to condemn are the hunters who, in the late 1880's, killed hundreds of elk, plus many deer and sheep, with little effort on their frequent organized hunts of "sport." Often the carcasses were left to rot or just the head trophy and a few choice steaks were taken from the fallen animals. The area around Estes Park was particularly noted for this irresponsible recreation. Actually the greatest reduction in animal numbers was accentuated by the actions of a few individuals and not by the concentrated efforts of all those present in the area. By 1913 elk had entirely disappeared from the Estes Park region and almost from the state. Also, during this infamous period sheep had been reduced from an estimated 4,000 in 1870 to a little over 1,000 forty years later. The mule deer, having a wider natural range, and not banding together in the summer as elk frequently do, were able to survive the hunting pressures somewhat better; however, the former abundant herds were then considerably reduced. The increase of settlements and the introduction of domestic stock further lessened the numbers by decreasing available feeding grounds.

The establishment of man in the main mountain valleys leading to the foothills was particularly important to the past and even present welfare of the rugged mountain sheep, one of the finest of all wild animals in the mountains. Wild sheep in early days always migrated through these valleys in the winter to the foothills (5,000 to 6,000 feet), where they were able to obtain bone-building minerals available in the sedimentary rocks. These minerals were not available in the high mountain granitic rocks and present

evidence indicates that they are highly essential for successful and sturdy lamb crops. Man's dwellings created an effective barrier to these animals and no longer did they make their way to the foothills each winter. This situation probably weakened the breeding stock. In addition, there were other declining factors such as hunting and in particular, the grazing of domestic sheep on former wild sheep ranges. This not only reduced available forage for wild sheep, but also introduced certain diseases of domestic sheep into the flocks. Sheep scabies reached epidemic proportions in the late 1800's, wiping out hundreds of wild bighorn.

A woefully miscast belief that gained common acceptance in the early 1900's forecast the decline of another group of interesting animals—the carnivores. The conception, fostered mainly by stockmen of the mountains, was that any kind of animal known to kill domestic sheep or cattle was detrimental and therefore all those animals should be eliminated. Consequently, an organized effort was made by stockmen with the cooperation of the federal government to "trap out" and kill these species. The success of their efforts in Colorado is evidenced in the virtual extinction of such indigenous mammals as the grizzly bear, the timber wolf, and the Canada lynx, and in a great reduction in the numbers of black bears, cougars, and bobcats. The coyote, while not so abundant in the mountains earlier, was the only predatory animal able to hold his own against the trapping, and has actually thrived in settled areas. Wildlife investigators have assembled a mass of evidence indicating that it is not the ordinary habit of these carnivores to feed regularly on domestic stock. Rather, it is the occasional or rare individual animal which confines its predation to domestic sheep or cattle. This being the case, it is the more usual practice now to confine extermination to those marauding **individuals**, rather than the entire race of animals. Many of the carnivorous animals have been protected since 1926 by state and federal laws. 9

The hopefulness of mankind regarding wildlife is seen in the aesthetic consideration finally given to these decimated animal populations. The establishment of protection areas, especially Rocky Mountain National Park in 1915, where **complete protection** is given to all animals, and the passing of state and federal laws for protection of game and regulation of hunting, were last minute efforts to save this splendid portion of American heritage.

In an effort to re-establish the elk in Rocky Mountain National Park, about thirty of these large animals were transported from Wyoming, 1913 to 1914, and released near Estes Park. Under complete protection these elk increased to approximately 1500 animals in 1941. Other early elk “releases” in Colorado brought the total elk population to about 25,000 in the state during the same period. The protection of mule deer in the Park increased their numbers to over 1,700 in 1941 and in Colorado to nearly 400,000. From less than 1,000 in number, the mountain sheep in the area started to increase gradually in 1909 and were “coming back” satisfactorily in the Park area. For Colorado in 1922, Seton optimistically estimated a bighorn sheep population of 8,000. However, in 1922 there began a gradual decline of sheep culminating in a counted number of 300 within the National Park boundaries in 1939. The estimate for Colorado in 1947 was 2,700, most of which were on National Forest land.

Of the formerly trapped smaller mammals in the region, the beaver and 10 marten have come back remarkably well, being abundant in many locations now. As an indication of beaver numbers and value in Colorado, some 40,000 beaver were estimated to be in the state in 1946; of these, 8,640 were trapped. A gross value of \$272,323 was realized from beaver for this year. The black bear, cougar, bobcat, red fox, and mink, while not considered common, are still sufficiently abundant to be glimpsed occasionally in the region of the Park. They appear to be maintaining their numbers. Coyotes, originally plains animals, are abundant, probably numbering over 200 individuals. The presence of other mammals such as skunk, badger, porcupine, marmot, muskrat, squirrel, rabbit, and other small rodents indicates they are holding a steady or increasing population, although no definite counts have been made. There is good reason to believe that the coyote, richardson ground squirrel, and abert squirrel are examples of mammals relatively new and increasing in the mountains over 6,000 feet.



Lush tundra vegetation, above 11,000 feet, provides summer forage for deer, elk, and sheep. Photo by author from Trail Ridge Road.



Elk wintering in lower mountain valleys. Cow elk in background is feeding on aspen bark. Photo by N.P.S.

WILDLIFE MANAGEMENT

The problem of restoring and maintaining wildlife in our national parks is not as simple as one might suppose. The mere creation of a preserve or area within which all wild animals are protected has proved non-sufficient. While it is true that these areas offer excellent opportunity for preserving a wonderful variety of primitive wildlife stock in its native habitat for future generations, it is equally true that the very laws establishing the National Park Service in 1916 have almost defeated their original intentions. The law emphasizes that the fundamental purpose of the Park Service shall be to conserve the scenery of Park areas and the wildlife therein, and to provide for the enjoyment of the same **in such a way as will leave them unimpaired for the enjoyment of future generations.** Roads, camp and picnic grounds, trails, and dwelling conveniences represent efforts by the Service to provide for the enjoyment of Park scenery and wildlife. However, each new road, trail, and “tourist convenience” removes wildlife food and cover from the Park and causes timid animals to retreat from these zones of human use. Moreover, it destroys the natural area which is supposed to be left “unimpaired for the enjoyment of future generations.” Furthermore, many animals lack the flexibility to adapt themselves to new habitats, once the ancestral areas are invaded or destroyed by man. With increasing numbers of visitors using the Parks, how can this tremendous conflict between man and wildlife be reconciled without the impairment of primitive wildlife or the restriction of human occupancy? It probably cannot be, and each ideal will have to sacrifice a portion of the original grand intentions. In every decision of human or wildlife use, the Park Service makes a strenuous effort to establish a happy medium and still conform to the basic purpose of the national parks.

The U. S. Forest Service which administers most of the mountain lands surrounding the National Park, considers wildlife as a **crop to be cultivated and harvested** by hunting under Colorado state game laws. National forest wildlife judgments are dove-tailed with other important forest policies such as timber growing, watershed protection, and domestic stock grazing.

The invasion of man into these now semi-wild areas has created changed, even severe, conditions for the existing wildlife. The efforts by man to counterbalance certain unfavorable conditions for the animals and therefore to conserve and administer them satisfactorily, constitute what is now 14 called wildlife management. Some of the problems that have existed and now exist in this area should be mentioned briefly to help us understand their scope and character.

ELK-RANGE PROBLEM

It is the normal habit of the elk in the region to spend the nine winter months in the lower mountain valleys and the summer months foraging on the fresh and succulent meadow grasses of the subalpine forest and alpine tundra country above 10,000 feet. Because of man-made restrictions in their winter territory, the elk congregate principally in Moraine Park, Beaver Meadows, and Horseshoe Park at this time. The past years of protection and adequate vegetation so increased their numbers that early in the 1940's there were some signs of large herds "grazing off" most of their natural winter food available in these valleys. To alleviate this overgrazing in the Park, the state of Colorado authorized elk hunting on adjacent national forest lands in 1941 in order to remove excess elk migrating out of the Park. However, this effort did not remove sufficient numbers from the Park herds. Instead of allowing this potentially serious condition to continue and the eventuality of either watching the elk starve or else feeding the animals year after year, a harvest of surplus numbers by hunting was conducted in the winter of 1944-1945. In this way a possible catastrophe of starving and dying off of the entire elk herd was halted. The present reduced elk herd of about 800 animals is considered more nearly within the winter range carrying capacity. The fact that there are no longer grizzly bear and sufficient cougar to take a normal number of these animals as food was an aiding factor to the rapid increase of elk. Concentrated numbers of elk seeking refuge in the aspen groves during heavy winters in the lower valleys have heavily damaged the aspen trunks by stripping bark for food. It is believed that pregnant cow elk, in particular, are able to obtain vitamin A from the aspen bark for their welfare at that time. This stripping or opening of the aspen trunk allows penetration of fatal tree

fungi, which may damage many aspen groves. Fortunately, the prolific aspen grows rapidly and soon should reappear satisfactorily.

DEER-RANGE PROBLEM

Essentially the same problem has existed for mule deer as for elk, with the exception that deer, which do not congregate so readily into herds, browse principally on low shrubs or bush plants rather than on grass. They do not strip bark from aspen trees. This habit permits both elk and deer to range fairly compatibly within the Park area. Nevertheless, the deer population was also considered to be excessively large for the amount of winter food available. Therefore, a smaller proportion of their number was also “removed” in the winter of 1944-1945, resulting in a present population of a little under 1,000 deer. For reasons unknown, however, the deer population has recently and gradually been declining within the Park. There is a possibility that the large number of coyotes now in the vicinity has assisted in keeping the deer herds from increasing.

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BIGHORN SHEEP POPULATION DECLINE

This country provides an extensive summer sheep range in the high rolling tundra and rugged peaks above timberline, in addition to a large wintering area in the lower timber and valleys. Strong winds in the winter sweep snow from the scant tundra vegetation and often make it possible for sheep to feed at these high altitudes even during the winter months. Even with these adequate topographic conditions, wild sheep in the National Park since 1922 have shown a slow, steady decrease in numbers until 1941, when there were about 300 sheep present. Since this date there has been a leveling off of sheep numbers, no decided increases or decreases being evident. All the related factors probably contributing to the decline of bighorn population or their present stability at low level are not known. One substantial reason advanced has been the deficiency of mineral in sheep diet in the higher mountains, as indicated on previous pages, with a resultant weakening of sheep stock and a consequent susceptibility to parasitism and diseases found prevalent among sickened and dead sheep over a period of years. Another

possibility for the decline may be present in the great increase of elk and subsequent competition for similar grass foods. The Park Service has placed salt and mineral blocks at known bighorn concentration places in an attempt to improve the physical condition of the sheep and thereby increase the sturdiness of their offspring. The results of this experiment are difficult to measure, but it is believed to have met with varying success.

BEAVER PROBLEM

The beaver, being a versatile and adaptable animal, is able to establish himself wherever there are small, permanent streams and sufficient aspen to provide him with logs and twigs for dams and houses and to provide food for his family. Consequently, any of the valleys in the Park which supply these requirements now contain numerous beaver. They represent more of a 16 nuisance factor than a real game management problem. Occasionally they will inundate and drown aspen stands and associated vegetation. Also, their dams will cause flooding of roads or other man-made improvements. Infrequently their dams are dynamited to release these waters and the beaver are live-trapped and transported to “wilder” areas in the state. Beavers were so numerous in the Park in 1941 that 106 were live-trapped and taken by state conservation officials to other Colorado areas. The fact that beavers work chiefly at night and have no serious predation worries has helped their normal increase.

These wildlife management problems are but samples of similar situations occurring throughout the country, but in varying degree and with different animals. These are types of conditions which wildlife managers must face. It is evident in the National Park that suitable study and research on such factors as animal-mineral requirements, parasites and diseases, bighorn-elk competition for food, rodent and big game food competition, condition and availability of winter foods, and predator relationships are vital to properly reconcile the use of the same area by man and various wildlife.

Animal populations are rarely in an “ideal condition of balance” in the same area. Rather, the normal condition is a series of population waves or fluctuations either increasing or decreasing the total numbers of a kind of animal. While some exhibit a kind of regularity, they do not always occur

with definite rhythm or in exact cycles. This was probably true in nature before the arrival of white man and will likely exist in wilder areas with little modification by man.

Another condition which must be considered normal among animals is the practice of predation, or killing of one kind of animal by another. The predator should be given the same opportunity to live its normal life as are the greatly favored species.

More often than not the predator takes the weakened or diseased animals of an area and thus aids in preventing the diseased animals from roaming among their fellows and spreading the ailment. Nature's sustaining law requires only the survival of the fittest and the predator fits admirably into this scene, unless he becomes too abundant.

The fear of wild carnivores or the "unknown" at night in the mountains is still somewhat prevalent. A comparatively brief knowledge of animal habits will soon force the less intrepid to concede that "wild animals" rarely attack a human in the wilderness, unless unduly provoked.

Finally, we should contemplate the wildlife of this country from another than the hunter or commercial aspect. The range limits of some of the more 17 superb animals in America today are shrinking into closely confined areas where the few spots of virgin wilderness remain. Man should direct his efforts toward assisting these grand animals to at least hold their own.

The thrill of close observation of a wild animal in natural surroundings, without the artificiality of bars or fences, is one of the outstanding satisfactions still available to man in this country. This inspiration and enjoyment, provided by the study and practice of wildlife preservation in the national parks, is of great importance as an intangible, but powerful influence on personal and national well-being.

LIFE ZONES AND ANIMAL DISTRIBUTION

Two interpretations governing the vertical distribution of plants and animals in the western mountain regions have been developed in the past years. Both are based on the premise that definite plants and animals (known as zone indicators) have maximum and minimum altitudes, above and below which they are unable to survive. The net effect is to group these plants and animals into belts or zones on mountain slopes, which vary but little in elevation above sea level throughout the western United States. The reasons why increases or decreases in mountain elevation so markedly affect the distribution of plant life, and to a much lesser degree the animal life, are closely correlated with the differences of temperature, available moisture, wind velocity, exposure of area to sunlight, soil, and topographic variations existing between these zones. Temperature in particular, being an easily measurable difference, has been used by Merriam in his classification of life zones. He computed the mean annual temperatures and made temperature summations for each clearly recognized zone of plant and animal life; he found that for each 1,000 foot rise in elevation there was a corresponding decrease in temperature of 3° F. Based on these temperature differences, definite geographical belts were formed and given names—arctic-alpine, hudsonian, canadian, transition and sonoran zones. Although in current use throughout the west, these zones are not clearly separable in the north-central Rocky Mountain region of Colorado, and therefore are not used here.

Weaver and Clements, following the same general idea, but considering all of the various factors mentioned above, devised a classification of zones which is applicable to the Park mountains and will be mentioned below. Actually, the trees and smaller plants fit very well into these zones, but animals, because of their mobility and wide adaptability, can hardly be classed in any definite zones. Most animals range at various times of the year through all three zones mentioned, but because a few do inhabit certain areas a large part of the time, they are considered to be typical of these zones. Probably the real limiting factor for animal localization is the degree of severe winter conditions they can endure; the more adaptable they are to low temperatures, the higher they may be found in the mountains throughout

the year. Of course, the distribution of herbivorous (plant-eating) animals largely determines the range of the predatory animals feeding on them.

LIFE ZONES (Weaver and Clements)

Alpine Zone—Any area above timberline—(About 11,300 feet) Grasses and herbaceous plants

These mammals could live the year 'round here if necessary, but all can and do range into the other two zones below:

Pika
Marmot
Pocket Gopher
Coyote
Red Fox
Snowshoe Hare
Mountain Sheep
Long-tailed Vole
Dwarf Vole

Subalpine Zone—9,000 feet to timberline—Dense forests of alpine fir and engelmann spruce, with occasional limber pine.

These animals extend but rarely into the alpine zone during the coldest part of the winter, and can and do range into the zone below:

Chickaree
Bobcat
Marten
Cottontail
White-tailed Jack Rabbit
Dusky Shrew
Golden-mantled Ground Squirrel
Least Chipmunk
Red-backed Vole
Porcupine

Long-tailed Weasel

Montane Zone—6,000 to 9,000 feet—Predominantly western yellow pine with scattered Douglas fir and aspen trees.

These animals are considered characteristic of this lowest Park zone and rarely wander into the subalpine zone.

Striped Skunk

Badger

Richardson

Abert Squirrel

Cliff Mouse

Ground Squirrel

All other mammals in the area, not mentioned above, probably range throughout these zones, especially during the summer months. Lodgepole pine may occur in the montane zone, while lodgepole pine and aspen are also abundant in the burned-over areas of the subalpine region. They are classified as sub-climax species and therefore not acceptable as zone indicators. When considering the altitude of timberline, it is important to understand that it will vary as much as 500 feet above or below the average of 11,300 feet, depending generally on the quantities of sunlight received. On warmer south and west slopes, timberline may go as high as 11,800 feet, while on the shaded north and east slopes it may drop down to 10,800 feet.

THE MAMMALS OF ROCKY MOUNTAIN NATIONAL PARK

While the term “animal” is commonly used in speaking of our four-footed wildlife, it is best to record with more complete accuracy that “animals” include any living thing having sensation and the power of voluntary movement. This would therefore admit a great variety of creatures such as one-celled protozoa, worms, fish, frogs, snakes, birds, and finally the four-footed animals mentioned—mammals. Mammals are set apart as a special group of animals for two reasons: they have some sort of hair covering on their bodies and the females are equipped with mammary (milk) glands for nursing their young, features which none of the other “animals” possess.

THE HOOFED ANIMALS

ELK (*Cervus canadensis nelsoni*)

Much taller and heavier than deer, with a dark brown, shaggy neck mane contrasting with the tan of the body. Large, round, cream-colored patch on rump. No antlers on females (cows). Running or galloping type gait.

A large number of these majestic animals are present in the region. In late June when snows melt from the high country meadows, bands of cows with their calves, may be found grazing in high valleys near timberline, or in the open tundra country above timberline. Cow elk usually bear a single calf each year. The characteristic white spotting on young calves usually disappears by mid-August, whereas deer fawn spots persist into the fall season. Occasionally, a bull will mingle and wander with a band. Large summer herds are often seen on the distant tundras from the Trail Ridge Road above timberline. Hikers have recently reported abundant elk in the extensive, isolated areas north of the Mummy range. The elk remain above 10,000 feet usually until the first week of September, when they migrate to

the lower timber and valleys. This is the start of the mating (or “rutting”) season, when the bull antlers are being polished and hardened. The challenging “bugle” of the bull elk can then be heard ringing out in a soul-stirring manner. The bulls at this time engage in a series of minor skirmishes with one another, for the purpose of dominating a group of cows (a harem) during the rutting season. Sometimes these meetings develop into 21 mighty battles, with these large, antlered beasts weighing up to 700 pounds apiece, pushing and gouging with their antlers and striking at each other with large front hoofs, until the vanquished flees. This is illustration on a grand scale, of nature’s way of providing the strongest animals for breeding and continuation of a strong stock.

Beaver Meadows and Horseshoe Park are particularly good places to view elk in the fall, from an auto. These cautious animals have excellent hearing ability and an exceptionally good sense of smell. They can detect a human a half mile away in proper wind, and once alarmed will retreat immediately to the wooded slopes. At the height of the rutting season, however, the elk are less easily alarmed. When elk can be seen from road parking areas, it is best to remain quietly in the car, as the gasoline odors seem to overpower any human scent they might obtain. Whatever the season, elk are most easily observed when they are feeding, either in early morning hours or at dusk. Often they can be “spotlighted” from the highway after twilight either on the tundra or in the valleys.

MULE DEER (*Odocoileus hemionus hemionus*)

A stout, chunky-bodied deer with a yellowish-gray coat, turning to gray in winter. Has big ears, small white rump patch; white tail with black tip is held down while running. Has stiff legged, bounding type gait. Antlers on males (bucks) only.

These beautiful creatures are the most abundant and widely distributed large animals in the Park. They may be found singly or in small groups throughout the forest and meadows, during the summer, and often graze at dusk and during the night near the Trail Ridge Road, from 8,000 to 12,000 feet altitude. In early June the females (does) usually bear their white-spotted, twin fawns in the deep forests; while the males (or bucks), having left the

family circle, are ranging far and wide in the wilderness. In early October the snows and winds usually drive the deer into the lower regions, where they assemble in small herds. The necks of the bucks begin to swell, heralding the approach of the rutting season, and a series of fights or “tussles” ensue among the bucks for possession of their harems of three to five does. These fights consist of the males horning and pushing one another around for short periods, when the stronger buck will finally throw the other off his feet and gore him with sharp, pointed antlers until he leaves. Mule deer herd together in the winter, feeding on aspen leaves and branches, and pawing away the snow from low bushes and shrubs to obtain their preferred diet. When the snow has melted on the steep south slopes in early June, 22 they break up into little bands and scatter to the four winds.

Mule deer have sharp eyes and a good sense of smell and hearing. However, they have a peculiar sense of curiosity and, if not alarmed, will often approach a spectator quite closely.

The number of points on mule deer antlers is a very poor indication of age. A yearling will usually have a pair of spikes six to eight inches long, but between two and five years of age the antlers may continue to hold the four points (tines). Deer (and elk) antlers frequently deteriorate with age and “go back” to two points or to a freakish number of points, sometimes numbering up to twenty-four points on a head. Very old deer and elk usually have short, scrubby sets of antlers and, of course, all elk and deer males lose their antlers in early spring and start immediately growing a new set. The hoofed animals in the Park are preyed on by cougar, coyotes, and bobcats. The coyote, originally a plains animal, has developed into a stronger and heavier mountain species, capable of bringing down adult deer and the younger elk and sheep.

MOUNTAIN SHEEP (*Ovis canadensis canadensis*)

A large, grayish-brown sheep with a distinct whitish rump patch. Males (rams) larger; up to 300 pounds, having horns which sweep back and down and finally, in older rams, curling forward. Females (ewes) weigh up to 175 pounds, with smaller horns pointing backward with slight curvature. Ewe horns have a vague

resemblance to the mountain goat horns, but there are no wild goats in the southern Rocky Mountains. Mountain sheep are also called bighorns.

No other animal of the Rockies is so symbolic of the wild, rugged grandeur of the Western mountain peaks as the mountain sheep. While they graze on sweet summer grasses and flowers of the alpine meadows and slopes, at 12,000 or more feet altitude, they are truly kings of all the vast domain they survey. They are all the more precious in the sanctuary of Rocky Mountain National Park. It is possible to drive up Trail Ridge Road and if one is ambitious, continue on foot up several miles of tundra slope to see one of the finest animal creatures placed on our planet. There are few places in this country where access to the high mountain peaks and sight of the bighorn is as easy.



Deer Fawn



Mountain Sheep Rams

The ewes bear their lambs singly, among the crags and rocky basins 25 high above timberline, in late spring. After a few weeks they congregate in small flocks along with the yearlings (and sometimes young rams) to spend the summer in thin-aired solitude. The older rams keep by themselves, alone or in smaller bands. When the winter winds and snows begin whirling around the lofty peaks, the sheep seek refuge in protected cliffs and timber, or even move to lower valleys. The mating or rutting season occurs in November, accompanied by terrific battles among the rams for their harems. The opponents race at each other, leaping into the air for the final, powerful crash of horns, which may be heard a mile away. After a number of such encounters, the smaller or weaker sheep gives up and walks away. The skulls of rams are well adapted to the terrible beating they take in battle. The top front of the skull is double, having a layer of bone, then a

space, then another layer of bone surrounding the brain case. In addition, the rams have a one-inch or more layer of shock-absorbing cartilage on the skull in back of the horns, joining the head and the backbone. Ram horns are not lost each year as are the antlers of deer and elk. Rather, they furnish a good indication of the age of the sheep, as they add a definite ridge or ring to the horn in its lengthening growth each fall season.

Bighorn bands have been observed recently in the following areas during the summer: The Never Summer Range, the Mummy Range, Flattop Mountain and peaks in vicinity, MacGregor Mountain, Specimen Mountain, Mount Ida and Sheep Rock, and on the crests near Trail Ridge Road above timberline. The small bands of sheep in the Park will shift with the season and with the year, but the last three named areas probably offer the easiest opportunity for viewing them. Sheep are usually on the move and feeding only in the very early morning hours and evening hours, often bedding down in secluded places in late morning and early afternoon. When stalking them, keep in mind that bighorn's eyes are exceptionally sharp and capable of detecting a moving human up to two miles away. If you can spot them first with a field glass and then keep out of sight until near them, your chances of a good view are much improved.

THE FLESH EATERS (CARNIVORES)

BLACK BEAR (*Ursus americanus*)

Bulky, heavily furred animal up to 3 feet in height when on all fours. Born with and retains either a black or cinnamon-brown fur. Adults weigh about 300 pounds, sometimes much more.

Although there are an estimated thirty black bear roaming the deep forests of the region, they are only occasionally seen because of their solitary, nocturnal habits. They are infrequently observed lumbering across a road or foraging an outdoor garbage pit in the evening. The latter practice is discouraged, when discovered, to prevent them from becoming "bum" bears. Because of their unpredictable and sometimes vicious manner, it is
unwise to feed or make friendly overtures toward any bear. They have

only fair eyesight, but in the woods can scent or hear a human coming long before he might be seen, and will slip silently away through the woods, despite their bulk. The heavy, clustered bear dung and large tracks are the most usual sign of bear in the region. The diet is largely ants, grubs, berries, roots, and some small rodents. Bears in the region will den up in early December and go into a light sleep or semi-hibernation, living off their stored fat layers. They may be easily wakened from this sleep. The females, which have mated the previous May, usually bear twin cubs in February. The cubs, strangely enough, are about the size of an adult squirrel when born. They grow rapidly and are soon out in the scattered snow fields feeding with Mama.

MOUNTAIN LION (*Felis concolor hippolestes*)

Very large, slender cat with small head and long, heavy, black-tipped, cylindrical tail. Fur soft, yellowish or reddish brown. Length, including tail, about 7 feet, height at shoulder almost 2½ feet, weight varies from 100 to 176 pounds.

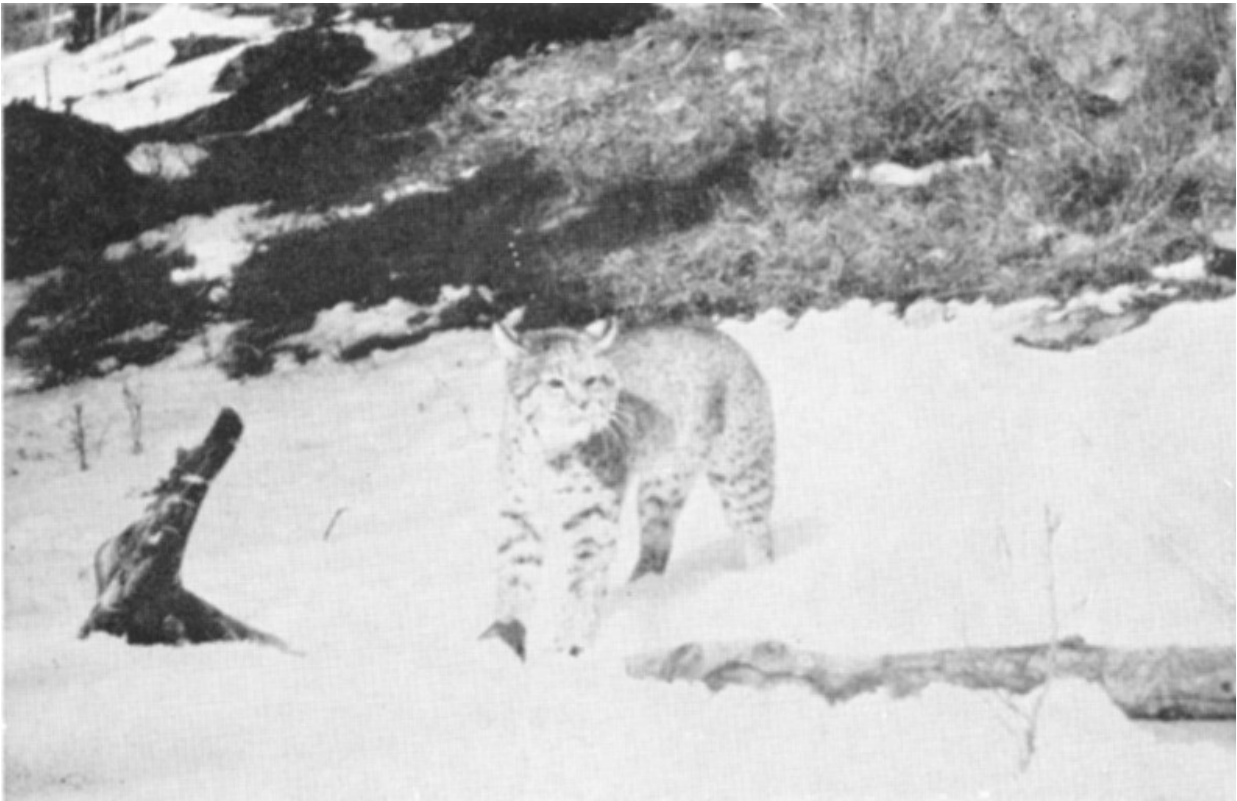
These great, sleek cats are among the most elusive of all animals to be seen in the wild. Because of their natural wariness and highly developed senses of smell and hearing, few persons have ever sighted the lithe, muscular body. Those who have, usually discover them from a distance, “sunning” on some rocky ledge or cliff. A few cougars are reported inhabiting the small canyons off the Devil’s Gulch area, northeast of Estes Park. If true, it is probably these cats making their circle “tour” of 50 to 100 miles in a few days’ search of game, that are infrequently seen in the Park. Cougars prefer fresh meat and prey chiefly on deer, but will catch rabbits and rodents occasionally. They have been known to trail a human long distances, but rarely show themselves or attack.

BOBCAT (*Lynx rufus uinta*)

General appearance like an extremely large domestic cat. There is considerable variation of color pattern in different kinds of bobcats, but the species seen in this area is buffy above with fine

streaks of gray and black; black bands appear prominently on legs. Total length about 3 feet; tail 6 inches. Weight up to 25 pounds. Note: The only animal the bobcat might be confused with is the lynx. The bobcat is smaller, buffy rather than gray, has smaller feet and short 1 inch ear tufts. The lynx is practically extinct in this area, while the bobcat or their tracks may be seen occasionally.

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Bobcat



Black Bear



WEASEL

Slender, brown with buffy underparts, black tip on tail; fur turns white in winter.



MINK

Dark brown fur and bushy tail, small ears; frequents stream areas.



MARTEN

Prominent ears, bushy tail, brown with yellow underparts; found in forest areas.



PIKA

Small, brown animal with short, round ears; no tail; found only above 10,000 feet, in rock piles.



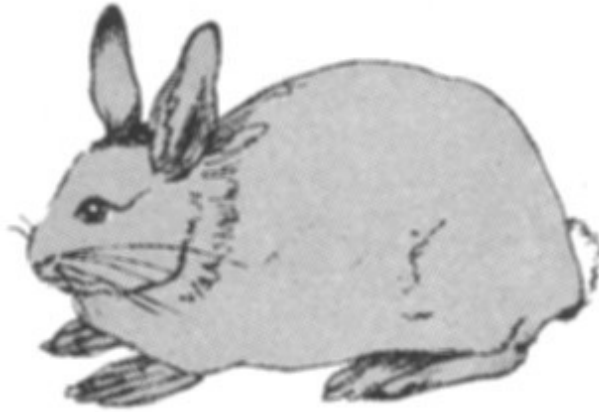
SNOWSHOE HARE

Smaller than a jack rabbit and with shorter ears; thick fur, gray in summer and pure white in winter; large hind feet.



JACK RABBIT

Very long ears, long hind legs; fur turns light gray in winter.



COTTONTAIL RABBIT

Smaller than hare and jack rabbit; feet and ears medium length; fur remains grayish-brown in winter.



CHICKA REE

Smaller grayish squirrel with white underparts, white eye ring, white fringe on tail; frequents spruce-fir forests.



ABERT SQUIRREL

Heavy bodied, long bushy tail, prominent ear tufts; fur is gray, brown or black; frequents yellow pine forests.



CHIPMUNK

Quick nervous movements; stripes on face and down **middle** of back, long tail, very common.



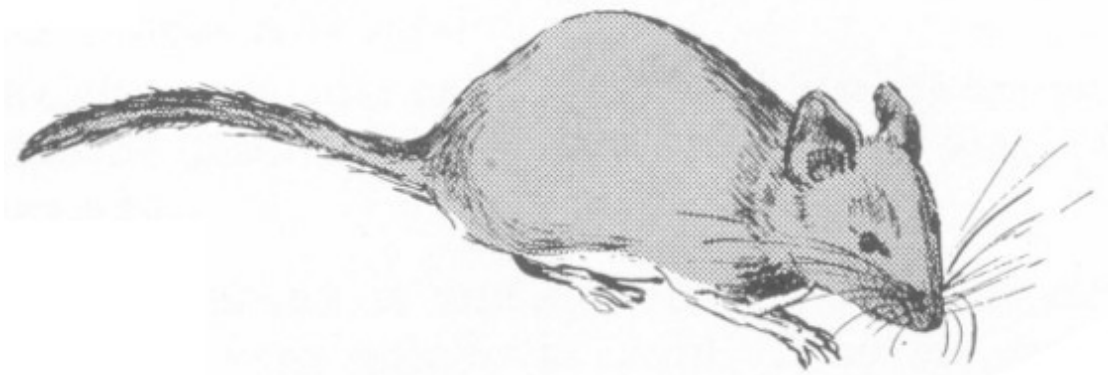
GOLDEN-MANTLED GROUND SQUIRREL

Larger than chipmunk; stripes **only on sides** of back; very common.



RICHARDSON GROUND SQUIRREL

Pale brown, short tail; often seen near highways in lower valleys.



PACK RAT

Large rat with brownish fur, bushy tail, and beady eyes.



POCKET GOPHER

Chunky, brown body, thick short tail, long front claws; seen near its earthen mounds.

The little bobcat ranges through the woods mostly at night seeking small rodents, rabbits, grouse, and ptarmigan. Like his giant cousin, the cougar, he will invariably detect quickly the presence of any intruder and quietly slip away. The presence of long hairs between his toes in winter, forming a “snowshoe-like” pad, enables him to travel swiftly through winter snows. Although wary of man, he will frequent settled areas where food in the form of rats, mice, and rabbits is common. 31

COYOTE (*Canis latrans lestes*)

Looks somewhat like a German shepherd dog with a yellowish gray coat and long, bushy tail. The coyote has a pointed nose, and a heavy tail which, when the animal is running, seems to float behind. Total length about 4 feet; weight up to 35 pounds. This species of coyote is usually larger than the familiar plains variety, and may be confused only with the larger wolf, which has disappeared from this region.

This crafty and bold “wild dog” is very common and increasing in the entire area, from the lower hills to above timberline. Their increase may be accounted for not only by their extreme cunning and adaptability to the invasion of man, but also because they produce the high average litter of six young each year. Scarcity of food, persecution by man, and the great stamina

of coyotes has helped him become the outstanding predator in North America, both in numbers and extent of range. They will eat practically anything—birds, insects, carrion, rodents, rabbits; and when in packs can overcome large game animals, which are in a weakened condition due to severe winters. I have seen coyotes in many of the lower valleys of the Park in mid-morning hours, “playing” with ground squirrels. They grab and fling them several times into the air, catching them expertly each time and finally gulping them down. The coyote becomes more awesome if you have heard its weird howl floating out of a moonlight night.

RED FOX (*Vulpes macroura*)

Reddish-gold coat and a long bushy white-tipped tail. Dark legs. Smaller than a coyote. Total length 3½ feet. Weight up to 14 pounds.

This fox is regarded as uncommon in the region and is difficult to see because it runs chiefly at night. They are swift and cunning, feeding on wood rats, mice, and birds throughout the area. Because of the value of their pelts in the fur trade, they have been heavily trapped and, not being as diversified in habit, have been unable to survive as well as the coyote.

32

CROSS FOX

This color variation of the red fox is similar except the coat is an intermixture of reddish, gray, and black tones. It has been seen in this region. The silver or black fox color phases of this red fox have not as yet, been reported for the Park. One litter of the red fox may contain several varieties of these phases.

BADGER (*Taxidae taxus taxus*)

Stout, flat-looking body with shaggy, silver-gray fur. Black and white distinctive markings on the face and head. Long, heavy claws. Total length about 28 inches. Weighs up to 20 pounds.

This compact, tough little badger, while more common in the plains and foothills, now digs its solitary burrow in the lower mountain meadows. As they capture prey by digging them out, they are usually found wherever there are ground squirrel colonies; but will also feed on skunks and marmots. They can dig themselves out of sight in the ground in a few minutes. Like the bears, they fatten up in the fall and go into a period of semi-hibernation from which they may waken and wander about during warmer winter days.

STRIPED SKUNK (*Mephitis mephitis varians*)

A stout bodied animal about the size of a house-cat, with a small head, large bushy tail, and short legs. Color black with a double stripe of white running the length of the back. Tail black and white. Total length about 28 inches. Weight up to 10 pounds.

This famous little night hunter sleeps most of the day and when awake is commonly seen roaming about human habitations. He feeds largely on small mice, insects, and also likes birds' eggs. He releases his potent scent only on extreme provocation or surprise and is actually quite a docile, friendly little fellow. If picked up by the tail, he **may or may not** fumigate the air.

SPOTTED SKUNK (*Spilogale tenuis*)

A smaller and more slender skunk distinguished by a number of narrow white stripes on the back which tend to break up, often resulting in spots. Rare in the Park and then only east of the Continental Divide.

MARTEN (*Martes caurina origenes*)

A large weasel-like animal with prominent ears and a bushy tail. Warm brown color except on chest and underparts which are yellowish. Total length about 25 inches.



Red Fox



Coyote



Marten

The elongated, agile-bodied marten is largely nocturnal, but because of his abundance is now rather commonly seen during the day in the subalpine forests of the Park. On the trails in Wild Basin, Bear Lake, and upper Colorado River Valley areas, he may be attracted to put in a bold appearance, by setting out a lure of smelly meat or fish. Ordinarily, they feed on chickarees and small rodents of the deep forest. They are primarily climbers, but are equally at home on the forest floor.

35

MINK (*Mustela vison energumenos*)

A slim, rich dark-brown animal with a pointed nose, small ears, and fairly bushy tail. Movements are snake-like. Does not turn

white in winter as will his smaller cousin, the weasel. Total length about 25 inches.

Aggressive and crafty killers, mink are infrequently seen along stream areas of the Park. They are as much at home in the water as out of it, catching fish and muskrats, as well as numerous small land rodents. Mink can travel miles along water courses with their bounding, graceful lope. Here they record their passage with tracks in the sand or mud. When angry, they emit a powerful, offensive odor.

LONG-TAILED WEASEL (*Mustela frenata nevadensis*)

Very slender weasel with a flattened head and beady eyes. Fur is dark brown, black tip on tail, and buffy underparts. Winter coat is snow white with black-tipped tail, and is then called "ermine." Total length 16 inches. There are about 36 different kinds of weasels in the United States.

It is incredible that such a small body could contain such a remarkably vicious nature as that of the weasel. Most animals kill for food, but the long saber-sharp teeth of the weasel kill wantonly and apparently just for the sake of killing. They first suck the warm blood from the base of the skull or neck of their victim and then eat portions of its meat and bones. They are quick and intelligent and can subdue animals several times their size. They are quite common throughout the Park up to timberline, and are so curious and unafraid that once seen, they may be attracted by making various squeaks and sounds.

SHORT-TAILED WEASEL (*Mustela streatori lepta*)

A very small weasel differing from the long-tailed weasel chiefly in size. Total length 9½ inches. Rare in the Park.

BEAVER (*Castor canadensis concisor*)

Compact, heavyset, water mammal with brown fur and a broad, horizontally-flattened, scaly tail. Large, webbed hind feet. Total length about 3½ feet. Average weight about 40 pounds. When swimming, only the top half of the head, shoulders, and part of the back appear above water. For positive identification, watch for the broad, black tail which may slap the water, or “flip up” when it dives.

This largest of North American rodents is very abundant and widely distributed in many of the mountain streams. To locate their dams, look for small pools or lakes in streams of heavily wooded sections. If new, the dams will be a mass of twigs and saplings carefully interlaced and sealed with mud; if old, the dams will be overgrown with grasses and small shrubs, but will still maintain the general shape and contour of a beaver dam. These dams will easily support the weight of a man. In the pond area or on the dam, a conical mass of mud and twigs, (the beaver lodge) some three to five feet high may be found, which contains the home of the beavers using that pond. Each lodge has an underwater entrance which is constantly in use, winter and summer. While beavers work mostly at night, it has been a regular practice in the Park to observe them swimming in their ponds just before nightfall. The Mill Creek, Hidden Valley, and Colorado River Valley areas have been especially good locations for sight of beaver. If aspen, which is both the beaver's food and construction material, have all been removed for a distance of five or six hundred feet from the pond, then probably the beavers have moved out and gone up or down stream to build a new pond. Muskrats may then occupy the entire pond.

MUSKRAT (*Ondatra zibethica osoyoosensis*)

This water mammal might well be a miniature beaver to the casual observer, with the one distinguishing feature of having a long, scaly tail flattened in the vertical plane instead of the beaver's broad, flat tail. Length not more than 2 feet. When swimming, only a small portion of the top of the animal shows above water, along with a thin edge of the tail, which is used with a sculling and rudder effect.

Muskrat are common in the Park, often living in beaver-made ponds. They are therefore often confused with beaver by the uninitiated, but if attention is given to the size and tail characteristics, there will be no identification difficulty. The muskrat or “rats,” as they are often called, build dens in the banks of the ponds and more rarely in this region, small grass and mud lodges. Their principal foods are rushes, grass, and water plants. In ponds containing active muskrat these plants are often found cut and floating near the banks.

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Weasel changing from brown summer coat to white winter fur



Photo by D. J. Obee

Badger



Porcupine



Muskrat

PORCUPINE (*Erethizon dorsatum epixanthum*)

39

Large, spiny rodent with high arched back, small black head, blunt nose, and heavy, short tail. Spines yellowish-white tipped with dark brown. Movements clumsy; slow, waddling gait. Total length up to 3 feet.

Common in the montane and subalpine forests of the region, the “quill pig” has been able to thrive because his potential predators can seldom discover that the only way to kill him, is to flip him over on his back and rip open his belly. Many interested animals, however, come away from a porky contact with a face or skin full of painful quills. Porcupine protect themselves by quick erection of masses of quills and by swift defensive swings of the spiny tail. These quills number up to 35,000 on a single animal. They spend most of their time clinging high in the tree branches, feeding on the green foliage

and the inner bark of pine trees. This accounts for the yellow “bare” patches sometimes seen on tree trunks. However, they have a strong predilection for anything containing even a fraction of salt; they will eat boots, axe handles, gun stocks, outhouse wood, and parts of buildings—anything where human perspiration has left a salty deposit. Porcupines, as well as smaller rodents, devour many elk and deer antlers left in the woods. They have a strange assortment of guttural sounds and cries which are sometimes heard at night. These uncanny noises emitting from the dark create interesting possibilities for more imaginative minds. The highly controversial question of porcupine mating is solved when we understand that the female has the muscular power of pulling in her quills closely to the body contour, permitting normal mating procedure. The single young is born in well-developed condition, but still enclosed in a membranous sack.

MARMOT (*Marmota flaviventris luteola*)

A medium sized western woodchuck having a variable shade of reddish-brown fur and a dark brown tail. Has a small band of white across the face. Total length up to 2½ feet and weight up to 20 pounds. They are 2 or 3 times larger than ground squirrels.

Marmots or “whistle pigs” as they are often called, are among the most easily observed mammals in the Rockies. They are found everywhere, particularly in rocky slides, boulder, and cliff areas from the Park boundary up to the tops of the highest peaks. Most of the parking area “feeding grounds” on the Trail Ridge Road contain their share of marmots, 40 which ordinarily feed on seeds, flowers, and grasses. They are a socially-minded animal, often living in small communities and posting a sentinel to give a high, shrill whistle at the first sign of danger. When sufficiently fattened in the fall, they find a snug hole in the rocks and drop off in the slow, deep sleep of the “true” hibernators.

ABERT SQUIRREL (*Sciurus aberti ferreus*)

A large, heavy-bodied tree squirrel with a long, bushy tail. Has 3 distinct color phases; gray, dark brown, and black, all of which are

common in the Park. Has long ear-tufts which are shed in the spring and grown out again by fall. Total length 20 inches.

Of the two kinds of tree squirrels in the Park, the aristocratic-appearing abert or tufted-ear squirrel is predominant in the montane or yellow pine valleys and ridges. They feed on the fine branches of yellow pine and on pine cone seeds. Their large, bulky nests of twigs and pine needles are placed high in the trees and are difficult to locate.

CHICKAREE (*Sciurus fremonti fremonti*)

A small, alert tree squirrel with back and sides a grayish-rust color; white underparts. Tail is white fringed. Distinguished from the abert squirrel by smaller size and the narrow white line around the eye. Total length is 13 inches. Formerly known as Fremont's squirrel.

This little chickaree or pine squirrel (as he is often called) is the chattering "alarm box" of the spruce-fir forests. They prefer the cool, denser, subalpine forests well above the range of the tufted-ear squirrel. Feeding chiefly on seeds of evergreen cones, they often leave great heaps of cone debris at their feeding places. The cones they bury for future use are often neglected and young seedling trees may thus spring up about their storage areas.

RICHARDSON GROUND SQUIRREL (*Citellus richardsonii elegans*)

A pale brown squirrel with a relatively short tail and a characteristically stiff, erect posture on occasion, which alone is enough for identification. Total length about 11 inches. Widely distributed in the west and formerly called Wyoming ground squirrel or picket-pin gopher in the southern Rockies. It is not a gopher.

These gregarious little fellows so frequently seen, alert and erect along the roadsides, have invaded the flat mountain meadows from the lower foothills. Living in colonies somewhat like prairie dogs (not found in this region),

their groups of small mounds are often scattered in the grass near fields that have been cultivated to hay crops.



Marmot



Chickaree



Golden-mantled Ground Squirrel

GOLDEN-MANTLED GROUND SQUIRREL (*Callospermophilus*

43

lateralis lateralis)

Larger than a chipmunk, chestnut-gray on head and back. Distinguished from chipmunks by having black and white strips on its sides (none down the middle of the back) and by having a plain face with no stripping. Not as nervous and quick as the chipmunk. Total length 11 inches. Formerly called Say's ground squirrel in this locality.

These trusting little fellows are the most commonly seen, photographed, and fed animals in the Park, (including the rangers). Dozens of these beautifully striped squirrels compete with chipmunks throughout the area for visitor “hand-outs” of peanuts. If left to themselves (those that haven’t forgotten how) they feed largely on plant material and seeds. There is a definite reason for this voracious engorging of food during the summer, which is enjoyed by the marmots and richardson ground squirrels as well. These animals are exemplary of the “true” hibernators of the animal kingdom. They go into burrows below frost line for a long winter period, curling up into tight little balls, and drop off into a death-like, torpid sleep from which they are aroused with some difficulty. Their temperature may drop from around ninety degrees F. to only forty; their heartbeat may reduce from approximately two hundred beats per minute to four or five; their oxygen consumption is less than ten per cent of the amount used in active condition. They utilize part of their stored-up summer fat for the little energy needed to keep alive. When the warm days of spring arrive, they dig their way out of the ground and search again for food and the customary “handouts.”

LEAST CHIPMUNK (*Eutamias minimus operarius*)

This little chipmunk is distinguished from the only other similar rodent in the Park, the golden-mantled ground squirrel, by its small size and nervous habit, stripes down the middle, as well as the sides of the back, and by narrow strips of black and white on its face. The tail is relatively long and bushy. Total length 8 inches. Found in Colorado east of the Continental Divide, from foothills to above timberline. Another least chipmunk (*Eutamias minimus consobrinus*) probably overlaps the range of *operarius* near the Continental Divide, and occupies the west half of the Park.

This quick nervous chipmunk is very common in all zones over 5,000 feet and is found scurrying among the rocks and along the forest floor, as well as running on tree trunks and branches. Like the ground squirrel, it has well-developed cheek pouches in which it can store an amazing amount of food. While it has a form of hibernation, it does not go into the deep torpor of the hibernating ground squirrel. It has been seen in the dead of winter running over the snows.

WESTERN CHIPMUNK (*Eutamias quadrivittatus quadrivittatus*)

A close relative of the least chipmunk of northwestern Colorado, which is rarely seen within the Park boundaries, and then not above 9,000 feet; is slightly larger and has a proportionately longer tail than the least chipmunk. Total length 8½-9½ inches.

NORTHERN POCKET GOPHER (*Thomomys talpoides fossor*)

A reddish-brown rodent with long, heavy, front digging claws. When compared to ground squirrels the pocket gopher has a heavier, chunky body and a shorter, thick tail. Has large, furlined cheek pouches. Total length 8½ inches. This group of pocket gophers is widely distributed from the great plains to the Pacific coast. Another subspecies (*Thomomys talpoides clusius*) occurs rarely in the Park although it is common in the plains and foothills. It resembles *fossor* except that its fur is brownish-gray.

Spending most of its life under the ground this well equipped excavator digs an amazing labyrinth of tunnels in western soils. It can make well over 200 feet of tunnel in a single night, usually digging down 5 or 6 feet below the surface. The prominent locator signs of the gopher are earthen mounds about 12 inches in diameter and 4 inches above ground level. In the center of the mound is an entrance to his tunnel system. In winter this tough, little rodent moves around above ground under the snow. His digging continues at this time and the excavated earth is pushed out of the ground and into his snow tunnels. When the snow melts the next spring the long, irregular earth cylinders lying on the ground are exposed. This shy, secretive gopher is difficult to see even at his earthen look-out mound, because he rarely exposes himself when intruders are nearby. They are found occasionally in meadows up to timberline.

BUSHY-TAILED PACK RAT (*Neotoma cinerea orolestes*)

A large rat, reddish-gray with heavy black linings on the back; color variations are considerable and may even grade into

yellowish-gray. Bushy, gray tail; large ears and beady eyes. Total length up to 17 inches. Formerly known as a wood rat.

45



Least Chipmunk

This notorious mountain rat is widely distributed over the state from 4,600 feet up to the tops of the highest peaks. Often frequenting buildings and cabins in the mountains, the pack rat has also been labeled a trade rat. These names are derived from the animal's habit of carrying off any loose article it may find and sometimes leaving other objects in the place of those taken. While these rats do possess a distinct musky odor, they, like many of the other rodents, have unduly suffered in character by comparison with the universally disliked house rat. A rodent ordinarily found in rocky places, they are much cleaner in appearance and habits than their dirty, disease-laden relatives of the city. They build a large, globular nest of soft, shredded materials and are most active in the first few hours of the night, and again before dawn. Usually they are very furtive.

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DEER MOUSE (*Peromyscus maniculatus rufinus*)

A round-eared mouse with tawny brown upper parts, and white feet and underparts. Total length up to 6 inches.

Like the pack rats, these are common in the mountain regions, particularly around dwellings, although they are much more easily seen than the rats. Unlike the disagreeable house mice, they keep themselves exceptionally clean. As they do not hibernate, they are commonly seen throughout the winter months from valley to timberline. They are also known as the white-footed deer mice of the mountains.

CLIFF MOUSE (*Peromyscus nasutus nasutus*)

A very large-eared mouse with dark, black and gray back. Total length up to 6 inches.

Sometimes called the long-nosed deer mouse, they are found occasionally in the Park, chiefly east of the Continental Divide. They prefer living in rocky areas, and are not as abundant as the true deer mouse (*maniculatus*).

JUMPING MOUSE (*Zapus princeps princeps*)

A large mouse with a dark back and greatly elongated hindlegs. Very long tail. Total length 9 inches.

This “kangaroo-like” mouse is found more commonly in the vegetation along the cold, rushing mountain streams. Although weighing less than an ounce, they make average hops or leaps from five to six feet at a bound.

THE VOLES

These little mammals are all members of a family of rodents which are set apart by their stocky, clumsy build, small ears, blunt heads, and certain skeletal differences. They have a decided preference for the colder

regions of the globe and are generally a populous group in the world of rodents, the most numerous and widely distributed of which are the meadow voles. They are merely listed here to show the variety that have been found in the Park and to give an indication of their relative abundance. The final identification of some of them is possible only on close examination and measurement. For identifying characters, consult the reference mentioned in the preface.

LONG-TAILED VOLE (*Microtus longicaudaus mordax*)

Common in all types of habitat.

DWARF VOLE (*Microtus montanus fusus*)

Common on grassy hillsides and drier meadows.

MEADOW VOLE (*Microtus pennsylvanicus modestus*)

Common in damp meadows.

RED-BACKED VOLE (*Clethrionomys gapperi galei*)

Occasional in damp woods.

NORTHERN VOLE (*Phenacomys intermedius intermedius*)

Rarely found in the subalpine zone.

THE HARES, RABBITS, AND PIKAS

These animals were formerly included in the order of rodents. However, instead of having the rodents' four front (incisor) teeth, (two above and two below), these have six. The extra pair are tiny and not very useful, being placed directly behind the upper front teeth. This anatomical difference is the scientific basis for separating the rodents and the rabbits.

PIKA (*Ochotona princeps saxatilis*)

A small, tailless member of the rabbit family, guinea-pig like in form, with short, round ears and a varying colored coat, ranging from buffy to brown. Total length 7 inches. Height at shoulder 3 inches.



Pika



Cottontail Rabbit

The little, abrupt-moving pika is found abundantly among the rock slides and slopes from timberline to the highest alpine peaks. Its high, quick bleat or shrill squeak, ventriloquistic in character, may be heard at any of the large rock piles in the alpine zone. Having a pronounced preference for cooler climate, it is but rarely found in the lower-valley montane zone. Wonderfully camouflaged to blend in with the rocks where it lives, it always crouches on all fours, never “sitting up” as do the similar sized ground squirrels. The pika spends his summer industriously gathering grasses and flowers and “curing” them on the sun-baked rocks to form his winter supply

of food or “haypile.” He does not hibernate, but lives actively throughout the winter, snug and secure from winter storms among the rock piles.

COTTONTAIL RABBIT (*Sylvilagus nuttallii pinetis*)

Small rabbit with feet and ears shorter than a jack rabbit or snowshoe hare. Fur remains dark grayish brown in winter. Short fluffy tail prominent while running. The snowshoe hare in its summer coat resembles the cottontail. However, the hare has larger hind feet and runs with great bounding leaps, in contrast to the short, rapid hops of the cottontail. Total length 16 inches. Ear length slightly over 2 inches.

The common little cottontail frequents the woods and valleys of the montane zone. The prolific mating habits are necessary to maintain their numbers, since they are heavily preyed on by many carnivorous animals. A single female may produce as many as 25 young during a year, in four or five matings. Cottontails, like the hares and jackrabbits, are subject to periodic diseases which may cause their virtual disappearance from a region. However, those rabbits surviving the epidemics will suddenly start to increase, and in a few years the area will again contain hundreds. These sudden changes in populations may occur regularly every five or ten years.

WHITE-TAILED JACK RABBIT (*Lepus townsendii townsendii*)

A large hare with very long ears (about 5 inches) and long, powerful hind limbs. Coat is a varying shade of gray turning paler in winter. In the very high altitudes, the coat will turn an almost white shade. in which coloration it resembles the snowshoe hare. Total length 24 inches. Found west of the Continental Divide. A subspecies (*Lepus townsendii campanius*) is found east of the Divide. Description same as above.

SNOWSHOE HARE (*Lepus americanus bairdii*)

Very much like the white-tailed jack rabbit, except the ears are not over 3 inches in length, and the coloration is more buffy-gray in summer. Changes to a thick, pure white coat in winter with only the tips of the ears remaining black. Total length 17 inches.

The snowshoe rabbit has gained its name from the hair covering the long toes and large feet. These “snowshoes” enable the animals to travel over the lightest snow crusts without sinking out of sight. Its wonderful protective coloration both in summer and winter, combined with a bounding jump which can carry it up to thirty miles an hour, provide this hare with means of evading some of its numerous predators. It is fairly common in the subalpine forests and tundra country both in summer and winter. The snowshoe hare has also been called the varying hare because of the molting or changing of coats with each winter and summer season.

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THE SHREWS (INSECTIVORES)

Shrews and moles both belong to the order of insectivores, so called because their diet consists principally of insects. No moles have been reported for the Park as yet. The nervous little shrew, smallest of all North American mammals, is distinctly mouse-like at first glance. Closer inspection, however, will reveal a very small, darting animal with long, pointed, quivering nose, tiny or hidden eyes and ears, a slender body, and a gray, velvet-like fur that brushes easily either way. Even the smallest shrews have razor-like teeth and vicious tempers, which give members of the mouse world great fear of these terrible little assassins. They will not hesitate to leap on fat mice twice their size and kill them with their tiny jaws. They live on the ground and are active during both day and night hours. Being difficult to observe, they are usually studied in the field by trapping techniques. Meadows, damp places, and rotted logs are favorite habitats. The kinds of shrews found in the Park are indicated below with their relative abundance. Complete identification may be made by reference to larger texts.

MASKED SHREW (*Sorex cinereus cinereus*)

Common in all life zones of the Park.

DUSKY SHREW (*Sorex obscurus obscurus*)

Common up to timberline in moist areas.

DWARF SHREW (*Sorex nanus*)

Probably occasional in montane zone.

WATER SHREW (*Sorex palustris navigator*)

Probably occasional in and about streams of montane and subalpine zones.

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Deer Mouse



Big Brown Bat

THE FLYING MAMMALS (BATS)

The bats are set apart from all other mammals because they have the power of flight. Otherwise, they possess the mammal characteristics of having a fur covering and suckling their young. Bats are rarely seen during the daylight hours, as they spend this time hanging upside down by claw-like feet in and about buildings and in branches of trees. They are mostly brownish or grayish with large “leathery” wings. Body length is from three to five inches; wing expanse six to twelve inches. Weight $\frac{1}{4}$ to 2 ounces. They swoop through the darkness with erratic, twisting movements and feed on flying insects. Their mouth contains a number of needle-sharp teeth. Bats have been greatly publicized because of their ability to fly through total darkness at relatively high speeds, avoiding all obstacles. In flight, the bats

utter cries too high-pitched to be audible to the human ear. The vibration of these cries are “bounced back” from objects as the bats approach, and are picked up by extremely sensitive ears in time to permit dodging the obstacle. In addition to these “silent” echo calls, they are capable of uttering staccato squeaks while in flight which are audible to the human ear. They will not strike or “hit” people in their vicinity. In winter they may go into a hibernation period in caves or migrate, like birds, to warmer climates. Very little is known about the species of bats existing in this mountain region, therefore the following list gives only an indication of the variety of bats believed to be in the Park.

BIG BROWN BAT (*Eptesicus fuscus fuscus*)

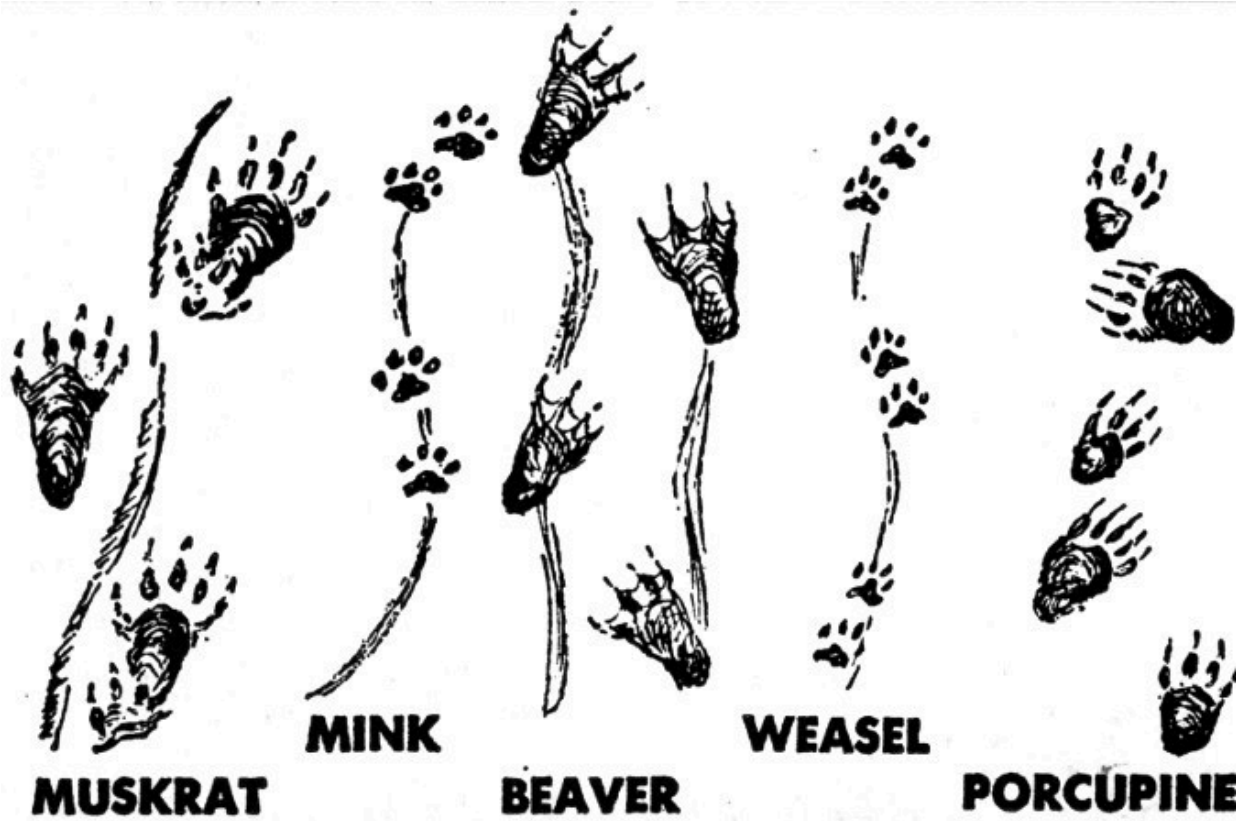
HOARY BAT (*Lasiurus cinereus*)

SILVER-HAIRED BAT (*Lasionycteris noctivagans*)

LUMP-NOSED BAT (*Corynorhinus rafinesquii pallescens*)

LARGE-EARED BROWN BAT (*Myotis evotis chrysonotus*)

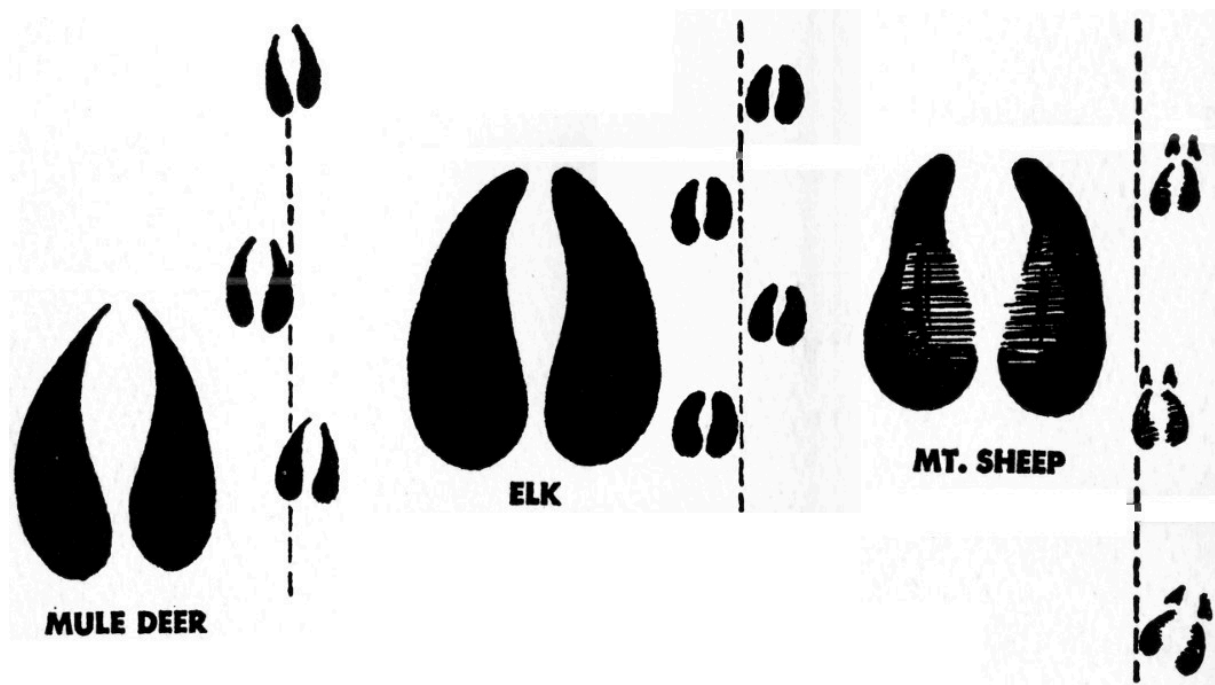
TRACKS



Tracks of muskrat, mink, beaver, weasel, and porcupine



Tracks of skunk, coyote, bobcat, and marten



Tracks of mule deer, elk, and mountain sheep



BLACK BEAR

COUGAR

Tracks of black bear and cougar

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OF THE ROCKIES ***

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